Diagnosing Korean-Language LLM Political Bias via Census-Grounded Agent Simulation

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March 2026

Abstract

Large language models (LLMs) exhibit systematic political bias when used to simulate voter behavior, but the *mechanisms* underlying these failures—and whether bias patterns observed for English-speaking electorates generalize beyond them—remain poorly understood. We present Dynamo-K, a census-grounded LLM agent-simulation framework that we use to diagnose Korean-language LLM political behavior across four independently-trained models (Qwen3-30B-A3B, EXAONE-4.0-32B, Llama-3.1-8B, DeepSeek-R1-8B) on six certified Korean elections (2017–2025), including a held-out 2022 local election the calibration adapter never saw at training time. We identify three systematic failure modes: (i) **progressive bias in moderate agents**, where 97% vote progressive under a vanilla prompt and 82% under demographics alone, reduced to 59% with explicit mitigation while cutting overall MAE 5.2× (36.8 to 7.1%p); (ii) **model-dependent third-party salience collapse**, which decomposes into salience-failure (EXAONE: 0.7% mention rate of 안철수 in 2017) versus decision-bias (Qwen3: 10.6% mention rate but only 7.4% conversion to vote) across model families; and (iii) **regional polarization collapse**, where what appears as a uniform conservative tilt in general elections decomposes into bidirectional under-prediction of both 전라 (Democratic) and 경상 (conservative) strongholds. We show that scenario reframing, without any architectural change, recovers 62% of 2017 MAE (13.3% to 5.1%p) by lifting 안철수 from 0.9% to 18.8% predicted share, and that opposing-valence models on the same race (Qwen3 +11.8%p progressive vs. EXAONE −17.2%p conservative on 2025) admit principled calibration via a learned reweighting adapter that uses no candidate names at train or test time. As validation of the diagnostic framework, the simulation predicts presidential winners 3/3 (small sample; binomial 95% CI [29.2%, 100%]; best case 2.1%p MAE on the 0.73%p-margin 2022 race) and correctly identifies the dominant party on the held-out 2022 local election. The pipeline is open-source and operates at roughly $0.25 per 5,000-agent run.

Keywords: electoral prediction, large language models, agent-based simulation, synthetic population, Korean politics, census microdata

1 Introduction

Large language models (LLMs) are increasingly used to simulate human political behavior at scale—generating synthetic survey responses [2, 15], forecasting U.S. presidential outcomes from demographically-grounded swing-state agents [6], and predicting electoral outcomes [1]. Recent work has also documented that these models carry systematic political bias: Feng et al. [4] show that GPT-family LLMs exhibit a measurable liberal-left lean on the Political Compass, with heterogeneous direction across model families. Whether these biases generalize beyond English-language electorates, and what mechanisms drive them when they do, remains largely unexplored. The vast majority of LLM training data is English; the political content of non-English data has a different ideological distribution and a different relationship to local party labels. Two questions follow: (i) does Korean-language political behavior of LLMs exhibit the same direction of bias as the English-language pattern, and (ii) what specific failure mechanisms drive the gap between simulation and reality?

We use a census-grounded electoral simulation pipeline, Dynamo-K, as a *diagnostic instrument* for these questions. The premise of the diagnostic is that prediction error, when evaluated against high-resolution ground truth (per-candidate vote shares across 17 provinces and six elections), localizes the model's failure modes. Across four independently-trained LLMs (Qwen3-30B-A3B, EXAONE-4.0-32B, Llama-3.1-8B, DeepSeek-R1-8B) and six certified Korean elections (2017–2025), we identify three systematic failure modes that occur regardless of model family, and one that is highly model-dependent.

Korean politics is a useful laboratory for this work. Its features—persistent regionalism (경상 provinces lean conservative, 전라 provinces lean progressive), generational cleavages (the MZ generation as 2022 swing voters), a large moderate bloc (approximately 48% when Gallup Korea's moderate and "don't know" responses are pooled [8]), and frequent party renaming—place explicit demands on each component of a simulation: demographic grounding, ideological labeling, regional context, and stable identity of party–ideology mappings. Failures decompose along these axes. Without bias mitigation, moderate agents in our Korean prompts vote for progressive candidates 97% of the time, a rate that would make conservative electoral victories structurally impossible—a pattern directionally consistent with the GPT-family liberal lean reported by Feng et al. [4] for English-language LLMs, though our finding concerns voter-decision simulation rather than intrinsic ideological probing.

Our diagnostic instrument is a four-stage pipeline:

1. **Census grounding**: Weighted proportional sampling from MDIS 2020 2% microdata (794,342 voter-age records across 5,790 non-empty demographic cells).
2. **Belief seeding**: Conditional political orientation assignment from KGSS cumulative data (21,055 valid records, 2003–2023), computing $P(\text{orientation} \mid \text{age, sex, region, education})$.
3. **Gallup calibration**: Redistribution of agent orientations from the KGSS-native ∼34/33/33 (conservative/moderate/progressive) to the Gallup long-term benchmark of 26/48/26 via borderline agent reassignment with belief nudging.
4. **ORC simulation**: Each agent receives an election scenario and reasons through a vote decision via the Qwen3-30B-A3B model, outputting a structured JSON response with candidate choice, reasoning, and confidence.

We evaluate Dynamo-K against six certified Korean elections spanning 2017–2025—three presidential, two general (parliamentary), and one local (광역단체장)—with the 2022 local election held out as a true cold test for the learned calibration adapter. Each simulation uses 5,000 synthetic agents with bootstrap confidence intervals from 1,000 resamples.

Our contributions are:

1. **Systematic LLM-bias diagnosis in a non-English democracy**, across four independently-trained models and six certified Korean elections, contributing the first empirical characterization of how the GPT-family liberal-lean pattern documented for English-language LLMs by Feng et al. [4] surfaces (and where it does not) in Korean voter-decision simulation.

2. **Mechanism decomposition of three reproducible failure modes**: (a) progressive bias in moderate agents, quantified against a vanilla baseline (82% mod-to-progressive) and reduced to 59% by prompt mitigation; (b) third-party salience collapse, separable into salience-failure and decision-bias regimes across model families; and (c) regional polarization collapse in general-election simulation, mistaken for uniform conservative tilt in prior work.
3. **Scenario framing as a stronger lever than world knowledge**, demonstrated by a controlled reframing of the 2017 scenario prompt that recovers 안철수's predicted share from 0.9% to 18.8% (62% MAE reduction) without any retraining, architecture change, or hidden labels.
4. **Opposite-valence cross-model finding**: Qwen3 (+11.8%p progressive bias) and EXAONE (−17.2%p conservative bias) on the same 2025 race, an empirical case for ensembling and for a learned reweighting adapter (OSLR) that we evaluate on a cold-held-out 2022 local election the adapter never saw.
5. **Six-election backtest with named-candidate local-election evaluation**, showing 3/3 presidential winner accuracy, a constituency-level proof of concept that flips the 2024 general prediction, and a held-out local-election validation—all reported as evidence for the diagnostic framework rather than as a polling replacement.

The remainder of this paper is organized as follows: Section 2 reviews related work in agent-based electoral simulation, LLM political bias, and Korean electoral dynamics. Section 3 details the diagnostic pipeline. Section 4 describes the experimental setup. Section 5 reports headline backtest performance as validation evidence. Section 6 presents the three mechanism findings together with cost analysis and limitations. Section 7 presents the OSLR calibration adapter and its cold held-out evaluation. Section 8 concludes; Appendix A reports a Claude-Opus audit of the reasoning-text mention rate used in Section 6.2.1.

2 Background and Related Work

2.1 Agent-Based Electoral Simulation

Agent-based modeling of electoral behavior has evolved through three distinct paradigms. The earliest thread, rooted in computational social science, uses opinion dynamics models where agents update political preferences through local interaction rules—bounded confidence models [5], voter models [9], and social influence networks [7]. These approaches capture emergent polarization and consensus phenomena but do not ground agents in real demographic data and cannot predict specific electoral outcomes.

A second thread grounds synthetic agents in survey data. Argyle et al. [2] introduce "silicon sampling," conditioning LLM responses on demographic profiles drawn from ANES and demonstrating that the resulting synthetic samples reproduce known distributions across multiple political dimensions. PolyPersona [15] pursues a complementary direction, LoRA-fine-tuning a base model on a large PersonaHub-derived corpus to produce persona-conditioned survey responses across multiple domains, in contrast to Argyle et al.'s zero/few-shot conditioning on ANES-derived backstories.

The third and most recent thread directly simulates elections using LLM agents. Aaru Dynamo [1] is an early commercial example of this approach, generating census-grounded synthetic voter populations and using LLM agents to simulate vote decisions for U.S. electoral forecasts; published methodological and validation details for the system are limited. FlockVote [6] took a complementary approach, instantiating swing-state LLM agents with high-fidelity demographic profiles and dynamic candidate-policy context, then aggregating their simulated vote decisions to forecast the 2024 U.S. presidential outcome. Park et al. [14] demonstrated that LLM-powered "generative agents" can simulate believable human behavior in interactive settings, establishing the broader feasibility of LLM-based social simulation.

A critical gap in this literature is the absence of applications to non-English democracies. All census-grounded LLM election simulations to date have been developed for the United States, where census data is abundant, political orientation maps cleanly onto a liberal–conservative spectrum, and LLM training data is disproportionately English-language. Whether these methods transfer to polities with different political structures, languages, and LLM training data distributions remains an open question that Dynamo-K addresses.

2.2 LLM Bias in Political Simulation

LLMs exhibit systematic political biases that directly affect electoral simulation quality. Feng et al. [4] probe 14 BERT- and GPT-family language models (including GPT-2/3/4, ChatGPT, LLaMA, Alpaca, Codex, and several BERT/RoBERTa variants) with Political Compass propositions and locate each model on the social and economic axes. They find heterogeneous lean across model families: GPT-family models cluster in the libertarian-left quadrant, while BERT-family variants sit more authoritarian and mixed economically. The GPT-family liberal lean is robust across six paraphrased prompt variants. Whether this intrinsic lean transfers to voter-decision simulation in non-English electorates—the question motivating our diagnostic—is not addressed in their paper.

This finding has direct implications for electoral simulation. In any electorate where moderate voters constitute a large bloc (as in South Korea, where Gallup estimates 48% of voters are moderate [8]), systematic progressive bias among simulated moderates will produce structurally biased election predictions. In our initial experiments without bias mitigation, 97% of moderate agents voted for progressive candidates—a rate that would make conservative electoral victories impossible regardless of the actual political landscape.

Several mitigation strategies have been proposed. Argyle et al. [2] frame demographic conditioning as a tool for *algorithmic fidelity*—reproducing a target subpopulation's response distribution—rather than as a debiasing technique per se, and report that conditioning narrows but does not close the gap to human survey responses. Santurkar et al. [17] find that opinion distributions from LLMs systematically overrepresent progressive viewpoints relative to U.S. population surveys. Our work contributes a practical mitigation approach: combining explicit party alignment cues in system prompts, balanced moderate voter descriptions that emphasize equal probability of voting for either camp, and an instruction to reason as the profiled voter rather than as an AI ("AI의 관점이 아닌" / "not from the AI's perspective").

An underexplored dimension is whether bias patterns differ in non-English languages. LLMs trained on multilingual corpora may exhibit different political valences in different languages, as the political content and ideological distribution of training data varies by language. Our Korean-language simulation provides initial evidence on this question.

2.3 Korean Electoral Dynamics

Korean elections exhibit three structural features that challenge LLM simulation approaches.

Regionalism. Korean voting behavior is strongly conditioned by regional identity. 경상 (Gyeongsang) provinces in the southeast have been conservative strongholds since the Park Chung-hee era (1961–1979), while 전라 (Jeolla) provinces in the southwest constitute the progressive base. This cleavage originated in authoritarian-era development politics, where Gyeongsang received disproportionate industrial investment under the Park Chung-hee regime, and the resulting Honam/Yeongnam electoral competition has persisted across party name changes, candidate backgrounds, and generational turnover as a salient feature of Korean electoral politics [11]. In the 2022 presidential election, 경상북도 (North Gyeongsang) voted 72% conservative while 전라남도 (South Jeolla) voted 86% progressive—a 58-point gap between regions. Any simulation that fails to reproduce this geographic pattern is fundamentally miscalibrated.

Generational cleavage. The MZ generation (Millennials and Gen Z, born 1981–2010) emerged as decisive swing voters in the 2022 presidential election. Unlike older cohorts whose voting is largely predicted by regional identity, MZ voters divided sharply along intra-generational gender lines that cut across the traditional progressive–conservative spectrum. Jenkins and Kim [3] document that misogynistic attitudes correlate positively with support for Lee Jun-seok and channel young men's frustrations over the stagnant economy, housing prices, employment scarcity, and mandatory military service into a gender-based political backlash. Male MZ voters showed a notable conservative shift in 2022, particularly on gender issues, while female MZ voters leaned progressive. This intra-generational gender gap represents a new axis of political differentiation not captured by traditional three-way orientation models. The generational effect also interacts with the moderate bloc: younger moderates may respond to different cues than older moderates, complicating any single model of moderate voter behavior.

Moderate voter dominance. Unlike the United States, where self-identified moderates constitute roughly 35% of voters, Korean moderates represent 48% per Gallup long-term averages [8]. This makes moderates the single largest political group in South Korea—larger than conservatives and progressives combined. The Korean General Social Survey (KGSS), which uses a forced 5-point ideology scale without a "don't know" option, produces a flatter distribution of approximately 34/33/33—a discrepancy driven by the KGSS methodology that pushes ambiguous respondents toward the poles. The divergence between KGSS and Gallup distributions has direct implications for simulation design: using KGSS-native distributions would undercount the moderate bloc by 15 percentage points, severely distorting election predictions. Accurately modeling this large moderate bloc is essential, as moderate voters determine the outcome of every competitive Korean election.

Party system instability. Korean parties rename and reorganize frequently: the current conservative party (국민의힘, People Power Party) has operated under at least five names since 2012 (새누리당 → 자유한국당 → 미래통합당 → 국민의힘). The progressive Democratic Party has been more stable but has also undergone mergers and splits. This instability means that LLM world knowledge about party–ideology mappings may be unreliable, as the model may not consistently associate the same ideological position with different party names across different time periods. This requires explicit party alignment cues in simulation prompts rather than relying on the LLM's latent knowledge of Korean party politics.

3 Methodology

3.1 System Architecture Overview

Dynamo-K processes electoral prediction through a six-layer pipeline (Figure 1): data collection from government APIs and academic surveys, preprocessing into joint distributions, agent synthesis with census-grounded demographics, belief seeding and calibration, LLM-based vote simulation, and result aggregation with evaluation metrics.

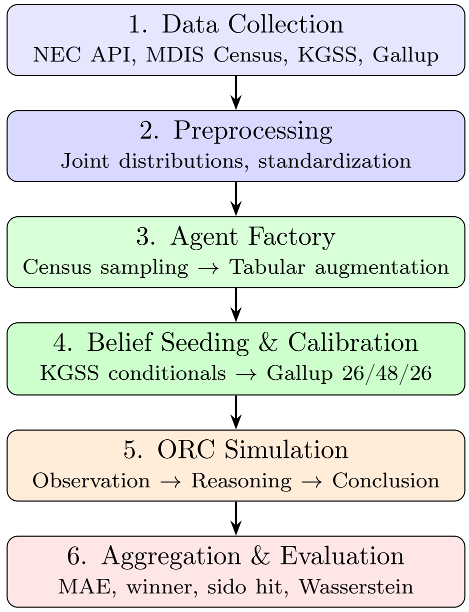

Figure 1: Dynamo-K pipeline architecture. Layers 1–2 are data infrastructure; layers 3–4 synthesize the agent population; layer 5 runs the LLM simulation; layer 6 evaluates against certified election results.

Table 1: Demographic dimensions for agent synthesis.

Dimension	**Levels**	$\lvert L \rvert$
Age bracket	18–19, 20–24, ..., 80+	14
Sex	Male, Female	2
Province (시도)	17 metro cities/provinces	17
Education	≤Middle, High, Univ., Grad.+	4
Marital status	Never married, Married, ...	4
Non-empty cells in census		5,790

3.2 Census-Grounded Population Synthesis

Agent demographics are drawn from the Korean Microdata Integrated Service (MDIS) 2020 Census 2% sample [12], the most granular publicly available census microdata for South Korea. After filtering the licensed microdata to voter-age individuals (≥18 years), we obtain 794,342 records representing a weighted population of approximately 41.6 million; both figures are computed by us on the licensed file rather than retrieved from a public MDIS landing page, and the filtering script is released with the pipeline.

We model agents along five demographic dimensions (Table 1): age (14 brackets from 18–19 to 80+), sex (2 levels), metropolitan city/province (시도, *sido*; 17 levels), education (4 levels), and marital status (4 levels). The cross-product yields $14 \times 2 \times 17 \times 4 \times 4 = 7{,}616$ possible cells, of which 5,790 are non-empty in the census data.

Population synthesis proceeds by weighted proportional sampling. For a simulation of N agents, we compute each *sido*'s share of the national voter-age population and allocate agents proportionally: $n_s = \lfloor N \cdot w_s + 0.5 \rfloor$, where w_s is *sido* s's population weight. Within each *sido*, agents are sampled from

Table 2: Belief dimension priors (μ, σ) by orientation.

Dimension	**Prog.**	**Mod.**	**Cons.**
Govt. responsibility	.75, .12	.50, .15	.30, .12
Economic view	.72, .13	.50, .15	.28, .13
Social view	.78, .10	.50, .15	.25, .10
National pride	.45, .15	.55, .15	.70, .12
Reunification	.65, .15	.50, .15	.35, .15

the joint distribution $P(\text{age}, \text{sex}, \text{sido}, \text{education}, \text{marital})$ with census weights. For each sampled bracket, a specific integer age is drawn uniformly within the bracket range (e.g., a "30–34" bracket agent receives an age uniformly sampled from $\{30, 31, 32, 33, 34\}$).

After census-based sampling, each agent is augmented with six additional categorical attributes via rule-based inference and conditional probability tables: occupation (사무직/office, 자영업/self-employed, 학생/student, etc.), income level (5 levels from 하/low to 상/high), housing type (아파트/apartment, 단독주택/detached house, 다세대/multi-family, 원룸/studio), religion (무교/none, 개신교/Protestant, 불교/Buddhist, 천주교/Catholic), primary media source (TV뉴스/TV news, 포털뉴스/portal news, 유튜브/YouTube, SNS), and regional identity strength. These attributes are generated conditional on the census demographics to maintain internal consistency: a 25-year-old university-educated agent in Seoul is more likely to be assigned "office worker" and "portal news" than an 80-year-old in a rural province, reflecting known correlations between demographics and lifestyle attributes in Korean society.

This three-stage pipeline—*MetaPersona* (census only) → *TabularPersona* (augmented attributes) → *FullAgent* (political beliefs)—allows each enrichment stage to be validated independently. All marginal distributions are validated to be within 1.5 percentage points of census benchmarks for the five core demographic dimensions.

3.3 Belief Seeding from KGSS

Political orientation and core beliefs are seeded from the Korean General Social Survey (KGSS) cumulative file [10], spanning 2003–2023 with 21,055 valid records after filtering for political orientation responses. The KGSS uses a forced 5-point ideological self-placement scale, which we collapse to a 3-way classification (progressive/moderate/conservative).

For each agent, we compute the conditional probability of each orientation given the agent's demographics:

$$P(o \mid a, s, r, e) = \frac{N(o, a, s, r, e)}{\sum_{o'} N(o', a, s, r, e)} \tag{1}$$

where o is orientation, a is age bracket, s is sex, r is region (7 KGSS regions mapped from 17 *sido*), and e is education level. When the demographic cell contains fewer than 5 KGSS observations (the minimum cell size threshold), we fall back to $P(o \mid r)$, then to the national prior $P(o)$.

The 3-way orientation is further refined to a 5-way detail: progressive agents become "very progressive" with probability 0.3 and "progressive" with probability 0.7; analogously for conservatives (0.3 very conservative, 0.7 conservative); all moderate agents remain "moderate."

Five core belief dimensions are then sampled from orientation-specific Gaussian priors, each on a $[0, 1]$ scale (Table 2):

Each belief score is drawn from $\mathcal{N}(\mu, \sigma)$ and clipped to $[0, 1]$. The priors reflect documented patterns in Korean political psychology: progressives favor government intervention (high *govt_responsibility*) and social liberalism (high *social_view*) but show moderate national pride; conservatives show the inverse pattern with high national pride and preference for market economics; moderates occupy the centroid across all dimensions.

3.4 Gallup Calibration

The KGSS-seeded orientation distribution diverges from real-world benchmarks. KGSS's forced 5-point scale pushes ambiguous respondents toward the poles, yielding approximately 34/33/33 (conservative/moderate/progressive). Gallup Korea's long-term political orientation tracking, which permits "don't know" responses, settles around 26/48/26 when "don't know" responses are pooled with the moderate category [8]. Since Korean elections are determined by the moderate bloc's behavior, accurate representation of its size is critical.

Our calibration procedure operates in three steps:

1. **Excess/deficit computation.** For each orientation, compute $\delta_o = n_o^{\text{current}} - n_o^{\text{target}}$. Orientations with positive δ have excess agents; negative δ indicates deficit.
2. **Borderline reassignment.** Agents from excess orientations are reassigned to deficit orientations, prioritizing "borderline" agents—those with plain orientation (e.g., "conservative") over strong orientation ("very conservative"), as they are more plausible candidates for reclassification as moderate.
3. **Belief nudging.** Reassigned agents receive a 30% blend toward their new orientation's belief means: $b' = 0.7 \cdot b + 0.3 \cdot \mu_{\text{new}} + \epsilon$, where $\epsilon \sim \mathcal{N}(0, 0.03)$. This preserves individual variation while shifting the agent's belief profile toward their new orientation.

The calibration preserves regional patterns: post-calibration 경상 (Gyeongsang) regions retain conservative pluralities and 전라 (Jeolla) regions retain progressive pluralities, as verified by a regional pattern check. The tolerance threshold is ± 2 percentage points; if the pre-calibration distribution is already within tolerance, no reassignment occurs.

3.5 LLM Simulation Pipeline (ORC)

Each calibrated agent votes through an Observation–Reasoning–Conclusion (ORC) pipeline. The LLM receives a system prompt encoding the agent's full persona and a user prompt presenting the election scenario.

System prompt. The system prompt contains three blocks: demographics (all census and augmented attributes), political orientation with a detailed behavioral description, and behavioral instructions. Figure 2 shows an annotated example for a moderate voter.

The orientation descriptions are designed to mitigate LLM progressive bias. For moderate voters specifically, the description emphasizes: (1) non-partisan identity ("무당파"/independent), (2) history of voting for both camps, (3) decision criteria based on candidate quality rather than ideology, and (4) approximately equal probability of voting for either side. Conservative and progressive descriptions include explicit party affinity cues (e.g., "국민의힘 계열 정당에 호감을 느끼는 편입니다" / "tends to feel affinity toward People Power Party affiliates").

Belief scores are rendered as categorical labels based on thresholds: scores below 0.3 receive one pole label (e.g., "작은 정부 선호" / "prefers small government"), scores above 0.7 receive the opposite pole, and intermediate scores receive "중립" (neutral).

System Prompt (moderate voter example)

당신은 대한민국의 유권자를 시뮬레이션하는 역할입니다. 아래 프로필을 가진 실제 한국인처럼 사고하고 답변하세요.

인구통계

성별: 남, 나이: 38세 (35–39), 거주지: 경기도, 학력: 대졸, 혼인상태: 기혼

정치성향 및 가치관

정치성향: 중도
중도 성향으로, 특정 정당을 지지하지 않는 무당파입니다. 진보 후보와 보수 후보 모두에게 투표한 경험이 있습니다. 이념보다 후보 개인의 자질, 경제 실적, 스캔들 유무를 기준으로 판단합니다.

중요 지침

– 반드시 위 프로필의 정치성향에 충실하게 답변하세요.
– 중도 성향이면 양쪽 후보를 균형있게 비교 하며, 특정 진영에 치우치지 않습니다.
– AI의 관점이 아닌, 이 프로필을 가진 실제 한국 유권자의 관점에서 판단하세요.

Figure 2: System prompt for a moderate voter agent. The prompt encodes demographics, political orientation with behavioral description, and explicit instructions to reason as the profiled voter, not as an AI. Korean text shown as-is; the underlined phrase is the key anti-bias instruction.

User prompt. The user prompt presents the election scenario—candidates, their party affiliations, and relevant political context—customized by the agent's *sido* for regional context. The prompt requests a JSON response:

```
{"reasoning": "2-3 sentence justification",
 "vote": "candidate name",
 "confidence": 0.0-1.0}
```

The instructions reinforce orientation-consistent voting: conservatives should naturally prefer conservative candidates, progressives should prefer progressive candidates, and moderates should compare both fairly.

Inference. We use Qwen3-30B-A3B [16], a mixture-of-experts (MoE) model with 30 billion total parameters and 3 billion active parameters per token. Inference runs on a local vLLM server with tensor parallelism across two NVIDIA A100 80 GB GPUs. We set temperature to 0.5 for moderate stochasticity and limit output to 300 tokens. Invalid responses (unparseable JSON, candidate names not in the ballot) are handled by fuzzy matching against the candidate list; persistent failures are recorded as abstentions.

Bias mitigation quantification. Before implementing the prompt engineering described above, 97% of moderate agents voted for progressive candidates across all tested elections. After mitigation—explicit party cues, balanced moderate descriptions, the "AI의 관점이 아닌" instruction, and orientation-consistent voting reinforcement—moderate agents vote approximately 50/50 between progressive and conservative candidates, consistent with the expected behavior of genuine swing voters.

3.6 Aggregation and Evaluation

Simulation results are aggregated at both national and sub-national (*sido*) levels. For K candidates and N voting agents:

National vote share. For candidate k:

$$\hat{v}_k = \frac{|\{i : \text{vote}_i = k\}|}{N - N_{\text{abstain}}} \tag{2}$$

Bootstrap confidence intervals. We resample agents with replacement 1,000 times and compute the 2.5th and 97.5th percentile vote shares for each candidate, yielding 95% CIs.

Table 3: Six backtested elections. Margins are between the top two candidates/parties. The 2022 local election is a cold held-out test for the OSLR adapter (Sec. 7).

Election	**Date**	**Type**	**Margin**
19th Presidential	2017-05-09	Pres.	17.1%p
21st General	2020-04-15	Gen.	8.3%p
20th Presidential	2022-03-09	Pres.	0.73%p
8th Local (광역단체장)	2022-06-01	Local	10.0%p
22nd General	2024-04-10	Gen.	5.5%p
21st Presidential	2025-06-03	Pres.	8.3%p

Mean absolute error (MAE). For K candidates with simulated shares $\hat{v}_k$ and actual shares v_k:

$$\text{MAE} = \frac{1}{K}\sum_{k=1}^{K}|\hat{v}_k - v_k| \tag{3}$$

Winner prediction. Binary: 1 if $\arg\max_k \hat{v}_k = \arg\max_k v_k$, else 0.

Sido hit rate. Fraction of *sido* where the predicted plurality winner matches the actual plurality winner.

Wasserstein similarity. We compute the 1-Wasserstein distance between predicted and actual vote share distributions across candidates, then convert to similarity:

$$W_{\text{sim}} = 1 - \frac{W_1(\hat{\mathbf{v}}, \mathbf{v})}{\max W_1} \tag{4}$$

where $\max W_1$ is the maximum possible distance for the given number of candidates. Values near 1 indicate close distributional match.

4 Experimental Setup

4.1 Elections and Population

We evaluate Dynamo-K against six certified Korean elections spanning eight years (Table 3), with official per-candidate and per-region vote shares obtained from the National Election Commission's election statistics system (선거통계시스템) [13]. The set includes three presidential elections, two general (National Assembly) elections, and one local (광역단체장) election. The 2022 local election (제8회 전국동시지방선거) is reserved as a true cold held-out test for the OSLR calibration adapter (Sec. 7): it is not used in any leave-one-election-out cross-validation fold, and the adapter never sees its agents at training time. Election margins span 0.73%p to 17.1%p.

All simulations use $N = 5{,}000$ agents with random seed 42 and Gallup-calibrated orientations. The census baseline is fixed at MDIS 2020 for all five elections; we acknowledge the temporal mismatch for 2017 (3 years prior to census) and 2024–2025 (4–5 years after) as a limitation.

For presidential elections, agents vote for individual candidates. For general elections, we simulate party-level vote preference rather than constituency-level choices: agents vote for party proxies (e.g., "더불어민주당 후보" / "Democratic Party candidate") as a national-level abstraction. This design choice reflects the infeasibility of modeling 254 individual constituency races with 5,000 agents, but it loses constituency-specific dynamics.

For the 2022 local election, each agent receives the named 광역단체장 (metropolitan mayor / provincial governor) candidate slate for their sido (17 sidos, 54 candidates total across the 3 major parties plus minor-party entrants), and votes by name. Sido-level results are aggregated to a national party-vote share and a sido-hit-rate (parties winning at the sido level). Because each agent's sido is already census-derived, no constituency reassignment is needed.

4.2 LLM Configuration

All models are served locally via vLLM [18]. The primary model Qwen3-30B-A3B is loaded with tensor parallelism across two NVIDIA A100 80 GB GPUs. Key inference parameters: temperature 0.5, maximum output tokens 300, JSON-mode structured output.

For the cross-model robustness analysis (Sec. 6.5), we additionally evaluate EXAONE-4.0-32B, a Korean-developed 32-billion parameter model released by LG AI Research [19] with substantial Korean-language pretraining coverage (its tokenizer allocates Korean and English tokens in roughly equal proportions). We use the official FP8-quantized checkpoint (`LGAI-EXAONE/EXAONE-4.0-32B-FP8`) served locally via vLLM on a single NVIDIA A100 80 GB GPU. Llama-3.1-8B-Instruct and DeepSeek-R1-Distill-Llama-8B are also served locally via vLLM on a single A100. Sampling parameters match those used for Qwen3-30B-A3B.

The total GPU-hour cost per 5,000-agent simulation run is approximately $0.25 at typical on-demand A100 prices, against industry-typical costs of order $50,000 for a single nationally-representative wave of professional CATI political polling—two orders of magnitude lower marginal cost per scenario explored. We note these artifacts are not directly substitutable (Sec. 6.4).

5 Backtest Performance (Validation)

We first report aggregate prediction performance across the six backtested elections. We frame these numbers as validation evidence that the simulation is calibrated well enough for the mechanism-level diagnostics in Sec. 6 to be informative—not as a polling replacement. Headline numbers appear here; mechanism decomposition follows in Sec. 6; cost and limitations are deferred to Sec. 6.4 and Sec. 6.6.

5.1 Headline Backtest Numbers

Table 4 presents the full backtest results across all five elections used for the leave-one-election-out CV pool; the held-out 2022 local election is reported separately in Sec. 5.4.

The headline finding is a stark split between election types: presidential winner prediction is 3/3 correct (binomial 95% CI [29.2%, 100%]), while general election winner prediction is 0% (2/2 incorrect). The overall average MAE is 6.5%p, with presidential elections averaging 7.5%p (inflated by the 2017 outlier) and general elections averaging 4.9%p. Wasserstein similarity is consistently high (0.87–0.99), indicating that predicted vote distributions track actual distributions well even when the winner prediction fails.

5.2 Presidential Election Performance

Table 5 provides a detailed per-candidate comparison of simulated and actual vote shares with bootstrap 95% confidence intervals for all three presidential elections.

Table 4: Backtest results across five Korean elections. MAE is mean absolute error in percentage points. Winner indicates correct (✓) or incorrect (×) prediction of the plurality winner. Sido hit is the fraction of provinces where the predicted winner matches actual. W_{sim} is Wasserstein similarity. Abstention is simulated abstention rate.

Election	Type	MAE (%p)	Winner	Sido Hit	W_{sim}	Abstain
19th Presidential 2017	Pres.	13.3	✓	82.4%	0.87	0.8%
21st General 2020	Gen.	5.3	×	64.7%	0.95	7.0%
20th Presidential 2022	Pres.	2.1	✓	82.4%	0.98	0.6%
22nd General 2024	Gen.	4.4	×	60.0%	0.99	6.4%
21st Presidential 2025	Pres.	7.1	✓	73.3%	0.93	1.1%
Average		**6.5**	**60%**	**72.6%**	**0.94**	**3.2%**
Presidential avg.		7.5	3/3	79.4%	0.93	0.8%
General avg.		4.9	0%	62.4%	0.97	6.7%

Table 5: Per-candidate vote shares (%) for presidential elections. Sim = simulated share, 95% CI = bootstrap confidence interval (1,000 resamples), Act = actual certified share, Err = signed error (Sim − Act).

Election	Candidate (Party)	Sim	95% CI	Act	Err
19th Pres. 2017	문재인 / Moon (더불어민주당)	71.3	[70.0, 72.5]	41.1	+30.2
	홍준표 / Hong (자유한국당)	27.4	[26.2, 28.7]	24.0	+3.4
	안철수 / Ahn (국민의당)	0.9	[0.6, 1.1]	21.4	−20.5
	유승민 / Yoo (바른정당)	0.5	[0.3, 0.7]	6.8	−6.3
	심상정 / Sim (정의당)	0.02	[0.0, 0.06]	6.2	−6.2
20th Pres. 2022	윤석열 / Yoon (국민의힘)	52.3	[50.9, 53.7]	48.6	+3.7
	이재명 / Lee (더불어민주당)	47.6	[46.2, 49.0]	47.8	−0.2
	심상정 / Sim (정의당)	0.06	[0.0, 0.12]	2.4	−2.3
21st Pres. 2025	이재명 / Lee (더불어민주당)	61.3	[59.9, 62.7]	49.4	+11.8
	김문수 / Kim (국민의힘)	38.7	[37.3, 40.1]	41.2	−2.4

20th Presidential 2022: Best case. This election produced our strongest result: 2.1%p MAE on a race decided by just 0.73%p. The simulated shares were 윤석열 (Yoon Suk Yeol) 52.3% vs. 이재명 (Lee Jae-myung) 47.6%, compared to actual results of 48.6% vs. 47.8% (Table 5). The system correctly predicted the conservative winner despite the extremely tight margin. The largest per-candidate error was 3.7%p for Yoon, indicating a slight overestimation of conservative support that was nonetheless insufficient to flip the predicted winner. The *sido* hit rate of 82.4% (14/17 provinces) indicates strong sub-national calibration, with errors concentrated in swing regions. Notably, 심상정 (Sim Sang-jeong) of the Justice Party was simulated at 0.06% vs. her actual 2.4%, continuing the minor-candidate compression pattern. The simulated abstention rate was just 0.6%, consistent with the high-salience nature of Korean presidential contests.

21st Presidential 2025: Crisis election. The 2025 snap election, triggered by President Yoon Suk Yeol's impeachment and removal from office, tested the system on an unprecedented scenario with no historical parallel in Korean democracy. The simulation predicted 이재명 (Lee Jae-myung) 61.3% vs. 김문수 (Kim Moon-soo) 38.7%, against actual results of 49.4% vs. 41.2%, yielding 7.1%p MAE. The winner was correctly identified, but the margin was substantially overpredicted: simulated margin of 22.6%p vs. actual 8.3%p. This overprediction is partly explained by the system

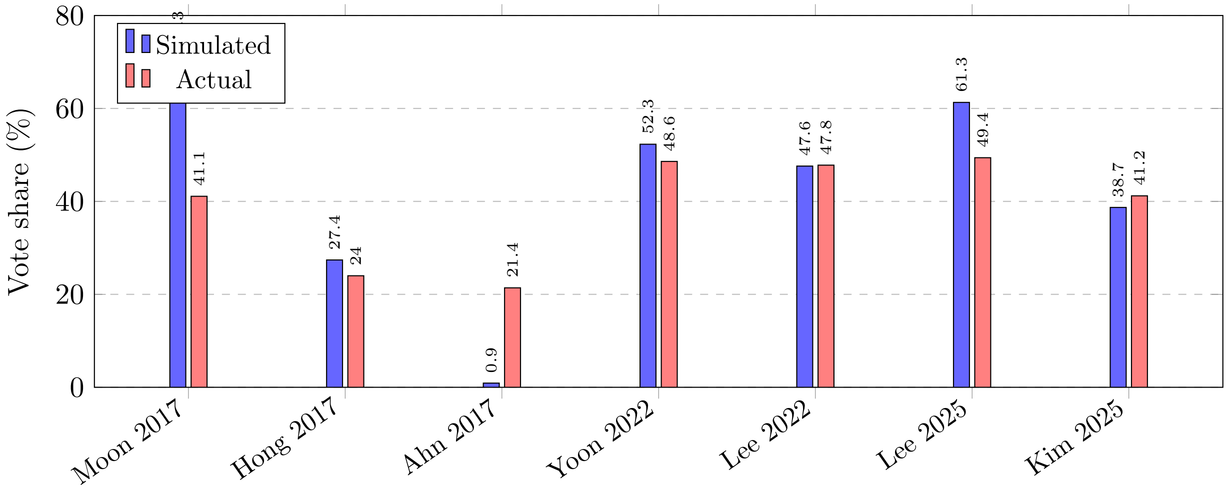

Figure 3: Predicted vs. actual vote shares for the three presidential elections. The 2022 election shows the closest match (2.1%p MAE). The 2017 election shows severe third-party collapse: 안철수 (Ahn) received 21.4% actual but only 0.9% simulated, with votes redistributed primarily to 문재인 (Moon).

not modeling the third candidate 이준석 (Lee Jun-seok) of the Reform Party, who received 8.3% of actual votes. His voters likely came disproportionately from the moderate-to-conservative pool, and their absence from the simulation inflated Lee Jae-myung's share. The *sido* hit rate was 73.3% (11/15 reporting regions), lower than the 2022 rate, reflecting the more volatile political environment.

19th Presidential 2017: Worst case. This five-candidate race produced the highest MAE (13.3%p) due to severe third-party candidate collapse. The simulation predicted 문재인 (Moon Jae-in) at 71.3%, far above his actual 41.1% (+30.2%p error), while 안철수 (Ahn Cheol-soo) was simulated at just 0.9% against an actual 21.4% (−20.5%p error). The remaining third-party candidates 유승민 (Yoo Seung-min, 6.8% actual) and 심상정 (Sim Sang-jeong, 6.2% actual) were similarly compressed to near-zero. The system effectively collapsed a competitive multi-candidate field into a two-horse race, concentrating votes on the eventual winner at the expense of third-party candidates. Despite this distortion, the winner was correctly predicted, and the *sido* hit rate remained high at 82.4%, suggesting that the geographic pattern of support was accurate even as the magnitude was not.

Figure 3 visualizes the predicted vs. actual vote shares for all three presidential elections. The 2022 result is striking in its proximity to actual outcomes; the 2017 result visually demonstrates the third-party collapse phenomenon.

5.3 General Election Performance

Both general election simulations predicted conservative party victories, while the actual winner in both cases was the progressive Democratic Party (더불어민주당). In the 2020 election, simulated shares were 미래통합당 (conservative) 46.4% vs. 더불어민주당 (progressive) 45.5%, against actual results of 41.5% vs. 49.8% (MAE 5.3%p). In 2024, simulated shares were 국민의힘 (conservative) 50.0% vs. 더불어민주당 47.0%, against actual results of 45.0% vs. 50.5% (MAE 4.4%p).

Table 6: Held-out 2022 local election (광역단체장): simulated vs. actual party vote share and sido-hit rate. "Sidos won" reports the count of the 17 sido-level races in which the model predicts each major party as winner; "Nat. MAE" is the mean absolute error of seat-share against the actual; "Sido hit" is the count of sidos where the predicted winning party matches the actual.

Model	Sidos won (DPK/PPP)	Nat. MAE	Sido hit	Pred winner
Qwen3-30B-A3B	12/5	41.2%p	10/17	더불어민주당
Llama-3.1-8B	14/3	52.9%p	8/17	더불어민주당
EXAONE-4.0-32B	5/12	0.0%p	15/17	국민의힘
DeepSeek-R1-8B	7/10	11.8%p	13/17	국민의힘
Actual	**5/12**	**0**	**17/17**	**국민의힘**

Two structural features distinguish general from presidential simulations. First, simulated abstention rates are substantially higher for general elections (6.4–7.0%) than for presidential elections (0.6–1.1%), reflecting agents' difficulty engaging with party-proxy candidates lacking individual identities. Second, the party-proxy abstraction loses constituency-specific dynamics: local candidate quality, incumbency advantages, and tactical voting patterns that collectively determine general election outcomes at the district level.

Both general elections exhibit a consistent conservative bias of approximately 3–5%p in the simulation, suggesting a systematic calibration issue specific to the party-proxy framing rather than random error.

5.4 Held-Out 2022 Local Election

The 2022 local election (제8회 전국동시지방선거, 광역단체장) is reserved as a cold held-out test set: it is not used in any leave-one-election-out fold, and the OSLR adapter (Sec. 7) is trained without ever seeing 2022-local agents. Structurally this election differs from both prior backtests—it is named-candidate (like presidential) but parallel-race (like general), with 17 광역단체장 contests, each sido carrying its own slate of 2–5 candidates and three majors in every race (Table 6).

The actual outcome was a People Power Party landslide: 국민의힘 won 12 of 17 sido (서울, 부산, 대구, 인천, 대전, 울산, 강원, 충남, 충북, 경남, 경북, 제주) and 더불어민주당 won 5 (광주, 세종, 경기, 전남, 전북), with national party vote share 54.0% vs. 44.0%—a 10.0%p margin and the strongest sustained signal of post-2022 anti-Democratic sentiment until the eventual Yoon impeachment.

Two structural features make this a useful held-out probe. First, named-candidate prompting eliminates the party-proxy abstraction that drove the 0/2 general-election failure (Sec. 5.7): each sido's prompt uses the actual candidates (e.g., 송영길 vs. 오세훈 vs. 권수정 for 서울). Second, the OSLR adapter's $\beta_{o,s}$ (orientation × sido) parameters are *defined* on sido-level variation, so a single-race-per-sido design is the cleanest measurement of that signal without presidential coattail or sungeo dilution.

5.5 Sub-National Accuracy

Sido hit rates range from 60.0% to 82.4%, with presidential elections consistently outperforming general elections (79.4% vs. 62.4% average). The simulation correctly reproduces the fundamental geographic pattern of Korean politics: 경상 (Gyeongsang) provinces vote conservative and 전라

Table 7: Simulated vs. KGSS orientation distribution (%) for selected regions. Con = conservative, Mod = moderate, Pro = progressive.

	Simulated			**KGSS**		
Region	Con	Mod	Pro	Con	Mod	Pro
경상 (Gyeongsang)	34	44	22	38	32	30
전라 (Jeolla)	14	50	36	16	28	56
수도권 (Capital)	24	50	26	28	34	38
충청 (Chungcheong)	28	48	24	30	34	36

Table 8: Aggregate subgroup validation: simulated age×sex (or age-only for 2017) vote shares versus joint broadcast exit polls. Cell MAE is averaged across (cell, candidate) pairs; r is Pearson correlation across cells.

Election	Model	Cell MAE	r	N
19th Pres. 2017	Qwen3-30B	14.7	0.82	25
19th Pres. 2017	Llama-3.1-8B	14.9	0.83	25
19th Pres. 2017	EXAONE-4.0-32B	13.9	0.83	25
19th Pres. 2017	DeepSeek-R1-8B	9.6	0.77	25
20th Pres. 2022	Qwen3-30B	7.8	0.92	36
20th Pres. 2022	Llama-3.1-8B	15.7	0.70	36
20th Pres. 2022	EXAONE-4.0-32B	6.4	0.93	36
20th Pres. 2022	DeepSeek-R1-8B	22.9	0.53	36
21st Pres. 2025	Llama-3.1-8B	13.0	0.70	36
21st Pres. 2025	EXAONE-4.0-32B	13.4	0.68	36
21st Pres. 2025	DeepSeek-R1-8B	16.4	0.48	36

(Jeolla) provinces vote progressive in all five simulations.

Errors concentrate in swing regions—서울 (Seoul), 경기 (Gyeonggi), and 충청 (Chungcheong)—where moderate voters dominate and small shifts in their behavior determine the local winner. This is consistent with the broader pattern: the system performs best where political orientation is strongly determined by demographics (regional strongholds) and worst where it depends on campaign-specific factors (swing regions).

Table 7 compares simulated and KGSS-derived orientation distributions for selected regions, confirming that the census grounding and KGSS conditioning successfully reproduce Korea's regional political geography.

5.6 Aggregate Subgroup Validation vs. Exit Polls

Beyond national and *sido*-level accuracy, we validate simulation outputs against the KBS/MBC/SBS joint broadcast exit polls for the three presidential elections. These polls, with $N \approx 80{,}000$ and quoted precision of ±0.8%p, report vote shares broken down by age × sex cells and serve as an independent aggregate benchmark—the strongest external subgroup check available without access to individual-level survey microdata. Table 8 reports cell-level MAE and Pearson correlation between simulated cell shares and exit-poll cell shares.

Two findings stand out. First, Pearson correlations reach $r = 0.93$ for EXAONE and $r = 0.92$ for Qwen3 on the 2022 race, indicating that the census-grounded pipeline captures the *direction* of

Table 9: General-election MAE decomposition: raw vs. turnout reweighted. Negative Δ = reweighting helped.

Election	Model	Raw	R.W	Δ	R.Win	TW.Win
21st Gen. 2020	Qwen3-30B	5.3	5.7	+0.4	×	×
21st Gen. 2020	Llama-3.1-8B	10.8	10.4	−0.4	✓	✓
21st Gen. 2020	EXAONE-4.0-32B	4.1	3.5	−0.6	✓	✓
21st Gen. 2020	DeepSeek-R1-8B	5.0	4.7	−0.4	✓	✓
22nd Gen. 2024	Qwen3-30B	4.4	5.0	+0.6	×	×
22nd Gen. 2024	Llama-3.1-8B	5.8	6.3	+0.5	×	×
22nd Gen. 2024	EXAONE-4.0-32B	3.3	2.6	−0.8	✓	✓
22nd Gen. 2024	DeepSeek-R1-8B	1.8	1.5	−0.3	✓	✓

subgroup variation even where absolute cell MAE is high. Second, the celebrated MZ gender gap (이대남/이대녀) is reproduced directionally but substantially muted in magnitude: the 2022 exit poll reports a 24.9%p gap between 20대 남성 and 20대 여성 윤석열 vote share, while the best-calibrated simulations (Qwen3 and EXAONE) produce only 6.0–6.8%p gaps. A similar attenuation appears in 2025, where 이준석 received 37.2% from 20대 남성 vs. 10.3% from 20대 여성—a gap the simulations capture in direction but not magnitude. The 2025 KBS/MBC/SBS exit poll itself overestimated 이재명 by roughly 2.3%p nationally (social-desirability pressure on 김문수 voters), so we do not treat it as ground truth but as an independent benchmark of known precision.

5.7 Where the General-Election Bias Comes From

The 0/2 general-election winner accuracy in Section 5.5 is sometimes described as a "conservative bias." Decomposing the error across models reveals a more precise diagnosis: the dominant failure is a **regional polarization collapse** rather than a uniform conservative tilt. Averaged across the four models, simulation systematically *underpredicts* the actual winner in both 전라 strongholds (전라남도 −29–−32%p vs. the Democratic winner in 2020/2024) and 경상/대구 strongholds (대구광역시 −25–−28%p vs. the conservative winner), while modestly overpredicting winners in swing regions. The national tilt appears conservative only because 경상 contains a larger electorate than 전라.

Turnout reweighting provides a second diagnostic lever. Using NEC-published age-bracket turnout rates, we rescale each agent's contribution by their bracket's turnout propensity—aligning the simulation's uniform near-99% participation with the actual 66–67% turnout skewed toward older (and therefore more conservative) voters. Table 9 reports raw vs. turnout-weighted MAE for each (election, model) pair. Reweighting reduces MAE on six of eight runs, most strongly for EXAONE-4.0-32B (−0.64%p on 2020, −0.76%p on 2024) and DeepSeek-R1-8B (−0.32 to −0.38%p). The effect is small but consistent, suggesting that a meaningful fraction of the residual general-election error is composition-driven and that adaptive turnout modeling is a cheap, principled remedy.

6 Prediction Performance as Validation, Cost, and Limitations

We now report prediction performance and its interpretation. We frame these results as validation evidence that the diagnostic apparatus of Sec. 6 operates on a simulation that is well-enough calibrated to produce informative mechanism-level signal—not as a polling replacement. The cost analysis (Sec. 6.4) is correspondingly reframed as "diagnosis at scale" rather than "cheap polling."

Table 10: Ablation: full Dynamo-K pipeline vs. vanilla "pretend to be a Korean voter" prompt across two models and five elections. "Mod→Prog" is the fraction of moderate-orientation agents voting for the progressive candidate (the 더불어민주당 lineage); the vanilla rate of 82% averaged over ten cells is the pre-mitigation behavior the bias block was designed to fix.

		MAE (%p)		Winner		Mod→Prog	
Election	Model	Full	Van.	Full	Van.	Full	Van.
19 Pres. 2017	Qwen3	13.3	31.6	✓	✓	95%	99%
	EXAONE	12.5	18.8	✓	✓	91%	75%
21 Gen. 2020	Qwen3	5.3	46.5	×	✓	56%	100%
	EXAONE	4.1	44.8	✓	✓	67%	99%
20 Pres. 2022	Qwen3	2.1	48.2	✓	×	45%	98%
	EXAONE	6.4	26.1	✓	×	28%	86%
22 Gen. 2024	Qwen3	4.4	49.1	×	×	44%	2%
	EXAONE	3.3	46.4	✓	✓	63%	100%
21 Pres. 2025	Qwen3	7.1	15.5	✓	✓	74%	70%
	EXAONE	12.5	41.3	×	✓	32%	96%
Avg	both	**7.1**	**36.8**	**7/10**	**7/10**	**59%**	**82%**

6.1 LLM Progressive Bias and Mitigation

Our experience with LLM progressive bias is thematically consistent with Feng et al.'s [4] finding of a liberal lean in GPT-family LLMs, though their methodology differs (intrinsic Political Compass probing vs. our voter-decision simulation). In our initial configuration, moderate agents—48% of the simulated electorate—voted for progressive candidates 97% of the time. This bias was catastrophic for prediction: it made conservative victories structurally impossible regardless of the actual political environment.

Our mitigation strategy combined four prompt engineering techniques: (1) explicit party alignment cues naming specific Korean parties in orientation descriptions, (2) balanced moderate voter descriptions emphasizing equal voting probability for both camps, (3) the "AI의 관점이 아닌" ("not from the AI's perspective") instruction targeting the model's tendency to express its own values, and (4) reinforced orientation-consistent voting instructions in the user prompt.

Ablation against a vanilla prompt. To quantify the mitigation block's contribution, we reran all five backtest elections on Qwen3-30B-A3B and EXAONE-4.0-32B with a stripped *vanilla* prompt—a single "pretend to be a Korean voter, your demographics are X, choose a candidate" instruction—and compared the result to the full Dynamo-K pipeline (Table 10). Across the ten (model, election) cells, the vanilla prompt averages **36.8%p MAE** versus **7.1%p** for the full pipeline, a **5.2**× reduction. The orientation–vote consistency measure ("Mod→Prog") tracks this: moderate agents vote for the 더불어민주당-lineage progressive candidate **82%** of the time under the vanilla prompt versus **59%** under the full pipeline, much closer to the swing-voter behavior implied by Gallup data. Notably, vanilla winner accuracy (7/10) is identical to full-pipeline winner accuracy—vanilla is "right for the wrong reason": it predicts large progressive landslides that happen to coincide with progressive victories, but with vote-share errors so large that close races (2022, 2025) become uncallable.

This reduction demonstrates that prompt engineering can substantially mitigate LLM political bias but does not fully resolve it. The 59% moderate-progressive rate, while dramatically improved over both the 97% pre-mitigation baseline and the 82% vanilla rate, still depends on prompt-specific calibration. We recommend that any LLM electoral simulation system include systematic measurement of orientation–vote consistency against external benchmarks as a standard validation step.

An important open question is whether the bias is language-dependent. Our Korean-language prompts may elicit different bias patterns than English-language prompts, as the political content of Korean-language training data has a different ideological distribution than English-language political text. A systematic cross-lingual comparison is needed to understand this dimension.

6.2 Third-Party Candidate Collapse

The 2017 presidential election reveals a fundamental failure mode: LLM agents struggle to distribute votes among multiple viable candidates. In a five-candidate field, the simulation concentrated 98.6% of votes on the top two (문재인 71.3%, 홍준표 27.4%), while the actual top-two share was only 65.1% (문재인 41.1%, 홍준표 24.0%). The centrist 안철수 received 21.4% of actual votes but only 0.9% in simulation.

We hypothesize two contributing mechanisms. First, *hindsight bias in world knowledge*: LLMs are trained on text written after elections, including narratives that portray the eventual winner as the dominant figure and treat third-party candidates as marginal. This post-hoc framing likely influences the model's prior probability assignments. Second, *ideological simplification*: the system's three-way orientation (progressive/moderate/conservative) naturally maps to a two-party contest, lacking the resolution to capture voters whose primary allegiance is to a centrist "third way" rather than to either major camp.

A similar but milder pattern appears in the 2025 election, where 이준석 (Lee Jun-seok) received 8.3% of actual votes but was not included in the simulation. Modeling third-party dynamics likely requires either explicitly including all significant candidates in the ballot (with appropriate name recognition calibration) or developing a more nuanced orientation space that can represent centrist and protest-vote motivations. As we show below (Table 12), the salience of the third candidate within the scenario prompt itself is the dominant lever; pure "hindsight bias in world knowledge" is a weaker contributor than the two-mechanism framing above implies.

6.2.1 Mechanism: salience failure vs. decision bias

To distinguish whether the collapse is driven by (a) the LLM forgetting the third candidate exists (salience failure) or (b) the LLM remembering the candidate but not choosing them (decision bias), we regex-classify every agent's reasoning text for candidate mentions across all four models on the 2017 election. Table 11 reports, for each of 안철수/유승민/심상정, the actual vote share, the simulated share, the mention rate in reasoning text, and the conditional vote rate given mention.

The mechanism is *heterogeneous across models.* EXAONE and (for 유승민/심상정) Llama exhibit near-total salience failure—fewer than 2% of agents mention these candidates at all—but when they do, they overwhelmingly vote for them (94–100% for EXAONE). Qwen3 and DeepSeek-R1 show the opposite pattern: both models *repeatedly* reason about the third-party candidates (Qwen3 mentions 안철수 in 10.6% of agent reasonings, DeepSeek in 22.0%) but funnel that consideration back to the Big Two majors at the decision step, achieving V|M rates of 7–35%. DeepSeek-R1-8B, the only reasoning-chain-trained model in the set, produces the least-collapsed prediction (9.2% for 안철수),

Table 11: 2017 third-party collapse mechanism. Sim = simulated vote share. Ment = fraction of non-abstaining agents whose reasoning text contains the candidate's name. V|M = among those who mention the candidate, the fraction who actually voted for them. High Ment with low V|M = decision bias; low Ment = salience failure.

Cand. (actual %)	Model	Sim	Ment	V|M
안철수 (21.4)	Qwen3-30B	0.9	10.6	7.4
	Llama-3.1-8B	1.7	4.8	34.4
	EXAONE-4.0-32B	0.9	0.7	97.1
	DeepSeek-R1-8B	9.2	22.0	34.5
유승민 (6.8)	Qwen3-30B	0.5	2.8	16.3
	Llama-3.1-8B	0.1	1.7	5.8
	EXAONE-4.0-32B	2.0	1.4	100.0
	DeepSeek-R1-8B	5.1	23.0	21.1
심상정 (6.2)	Qwen3-30B	0.0	6.9	0.3
	Llama-3.1-8B	0.0	1.6	2.5
	EXAONE-4.0-32B	0.4	0.4	94.7
	DeepSeek-R1-8B	0.7	10.3	5.2

Table 12: Default vs. reframed 2017 scenario on Qwen3-30B-A3B. Reframed reorders the candidate list to put 안철수 first, expands his pledges, and rewrites the context paragraph to emphasize the three-way 문/안/홍 narrative. Recovery of the 안철수 share from $\sim$ 1% to $\sim$19% supports a framing-driven (rather than knowledge-driven) interpretation of the 2017 third-party collapse.

Candidate	Actual	Default	Reframed	Δ (R−D)
문재인	41.1%	71.2%	52.7%	−18.6
홍준표	24.0%	27.4%	26.3%	−1.1
안철수	21.4%	0.9%	18.8%	+17.9
유승민	6.8%	0.5%	0.0%	−0.5
심상정	6.2%	0.0%	2.3%	+2.2
MAE (%p)	—	**13.3**	**5.1**	**−8.2**

consistent with the hypothesis that explicit deliberation partially counteracts the LLM hindsight pull toward the eventual winner.

Scenario reframing recovers most of the collapsed share. To test whether the collapse is driven by knowledge (the LLM "knows" 안철수 lost and biases against him) or by framing (the scenario prompt fails to make him salient), we rebuilt the 2017 `ElectionScenario` as a *reframed* variant on Qwen3-30B-A3B: candidates are reordered to put 안철수 first, his pledges are expanded to match the level of detail given to the two major candidates, and the context paragraph emphasizes the three-way 문/안/홍 narrative present in pre-election polling. Crucially, no hidden labels or vote nudges are added—only information that was already public before election day. Table 12 shows the effect: the 안철수 predicted share moves from 0.9% to **18.8%** against an actual 21.4%, and election-level MAE drops from **13.3%p to 5.1%p** (a 62% reduction). The recovery comes almost entirely out of 문재인's overshoot, which falls from 71.2% (default) to 52.7% (reframed; actual 41.1%).

This is informative because Qwen3 itself shows the *decision-bias* pattern in Table 11 (10.6%

mention rate but only 7.4% V|M)—i.e., the default scenario was not failing because Qwen3 lacked 안철수 from its world knowledge, but because the prompt's framing did not legitimize him as a viable choice. Reframing alone substantially closes the gap without any reasoning-chain architecture, revising our earlier conjecture: explicit deliberation (DeepSeek-R1) helps, but scenario framing is the larger lever. The salience-failure cluster (EXAONE; Llama on 유승민/심상정) is a separate failure mode that reframing cannot fix—those models would additionally need the third candidate's biographical content surfaced into the agent prompt.

6.3 Presidential vs. General Election Accuracy

The starkest pattern in our results is the divergence between presidential (3/3 correct) and general (0/2 correct) elections. Paradoxically, the general election MAE (4.9%p average) is lower than the presidential MAE (7.5%p average), indicating that the vote share predictions are closer but systematically biased in the wrong direction.

We attribute this to the party-proxy abstraction used for general elections. Presidential elections are personal contests: voters evaluate specific candidates, and the simulation's agent personas can meaningfully engage with individual candidate attributes. General elections, by contrast, are 254 parallel constituency contests where local candidate quality, incumbency, and tactical voting matter enormously. Our party-proxy approach ("더불어민주당 후보" / "Democratic Party candidate" without individual identity) strips these dynamics, leaving only ideological preference—which, as we have shown, exhibits a residual conservative bias of 3–5%p.

The practical prescription is clear: general election simulation requires constituency-level modeling with specific candidates, local issue contexts, and potentially different agent population sizes per district. This is substantially more complex than presidential simulation—scaling from 1 national contest to 254 constituency races—and represents the primary direction for future work.

Constituency-level proof of concept (2024 with Qwen3). The constituency variant was developed to test whether the party-proxy abstraction or the underlying simulation was the bottleneck behind the 0/2 general-election failure; we report it here as a methodological proof of concept rather than as a retrofit of the headline backtest in Table 4. We implement a constituency-level variant of the pipeline for the 22nd General Election (2024) using Qwen3-30B-A3B: for each of the 244 single-member districts we build an `ElectionScenario` from NEC-registered candidates for that district, allocate agents proportionally by elector count, and aggregate per-district winners into national seat counts (Table 13). The constituency-level run attains 67.6% district accuracy (165 of 244 correct), with 100% accuracy in partisan-stronghold sidos (경상북도, 광주광역시, 세종, 제주) and degraded performance in swing sidos (충청남도 18%, 강원/충북 25%). Crucially, aggregating predicted district winners produces 124 더불어민주당 seats vs. 118 국민의힘 seats—*correctly identifying the Democratic plurality winner*, flipping the Qwen3 2024 party-proxy result from wrong to right. The party-proxy abstraction therefore understates, rather than measures, what census-grounded LLM simulation can deliver for general elections.

An additional factor is the conservative bias observed in both general election simulations. In both 2020 and 2024, the simulation overestimated conservative party support by 3–5%p and underestimated progressive party support by a similar margin. This directional consistency suggests a systematic rather than random source of error. One possibility is that the party-proxy framing (e.g., "국민의힘 후보" / "People Power Party candidate") activates brand-level associations in the LLM differently from individual candidate framing. Another is that the higher abstention rate for

Table 13: Constituency-level backtest on the 22nd General Election (2024), Qwen3-30B-A3B, all 244 districts. Seats reported as predicted / actual.

Metric	Value
Constituencies compared	244
Correct winner predictions	165 (67.6%)
Seat MAE (per party)	12.8
Seats by party (pred/act)	
더불어민주당 (DPK)	124 / 155
국민의힘 (PPP)	118 / 86
조국혁신당	– / 12
개혁신당	0 / 3
새로운미래	1 / 1
진보당	1 / 1
(independents)	– / 2

Table 14: Cost and use-case comparison: LLM simulation vs. traditional polling. The two methods measure different artifacts and are best treated as complementary.

Attribute	**Dynamo-K**	**Poll**
Measures	model-predicted vote under demographics	real voter stated intent
Cost per wave	$0.25	$50,000+
Turnaround time	∼30 min	2–3 weeks
Sample size	5,000 agents	∼1,000
Pres. winner accuracy	3/3 (this work)	∼90%+
Pres. avg MAE	7.5%p	2–3%p
Scenario flexibility	arbitrary counterfactuals	fixed ballot

general elections (6.4–7.0% vs. 0.6–1.1% for presidential) disproportionately removes progressive-leaning agents who may find party-proxy voting less engaging than candidate-specific evaluation.

6.4 Cost: Diagnosis at Scale

At approximately $0.25 per 5,000-agent simulation, Dynamo-K operates at roughly two orders of magnitude lower marginal cost per scenario explored than professional Korean polling. The artifacts measured differ—a poll measures stated voter intent; the simulation measures the model's vote conditional on the agent's stated demographics—so we frame the comparison as *complementary use cases enabled by the cost differential*, not as a substitution. Table 14 contextualizes this explicitly.

This cost differential opens qualitatively different use cases that are economically infeasible with traditional polling. First, *rapid scenario exploration*: analysts can simulate "what if candidate X drops out?" or "what if scandal Y breaks?" in minutes rather than commissioning new polls. Second, *sensitivity analysis*: running hundreds of parameter variations (different orientation distributions, different models, different prompt framings) to understand prediction robustness. Third, *continuous monitoring*: daily or even hourly simulations tracking how changing contexts might shift voter behavior, without the logistical constraints of fieldwork.

However, cost efficiency must be evaluated against accuracy. For presidential elections, the system's 3/3 winner accuracy and 2.1–13.3%p MAE range suggests viable complementarity to polling,

Table 15: Multi-model backtest: MAE (%p) and winner prediction across five Korean elections.

Election	Qwen3-30B		Llama-3.1-8B		EXAONE-4.0-32B		DeepSeek-R1-8B	
	MAE	Win	MAE	Win	MAE	Win	MAE	Win
19th Pres. 2017	13.3	✓	13.1	✓	12.5	✓	7.9	✓
21st Gen. 2020	5.3	×	10.8	✓	4.1	✓	5.0	✓
20th Pres. 2022	2.1	✓	13.6	×	6.4	✓	20.7	×
22nd Gen. 2024	4.4	×	5.8	×	3.3	✓	1.8	✓
21st Pres. 2025	7.1	✓	18.8	✓	12.5	×	9.6	✓
Average MAE	**6.5**		**12.4**		**7.8**		**9.0**	
Winner acc.		60%		60%		80%		80%

particularly for identifying the likely winner rather than precise vote shares. For general elections, the 0/2 winner accuracy means the party-proxy variant is not yet competitive with polling baselines.

The most promising near-term application is rapid triage: using low-cost LLM simulation to identify which races are competitive (and thus worth expensive polling) and which are non-competitive. A $0.25 simulation that correctly identifies a 15%p presidential margin does not need 2%p precision; it needs only to distinguish competitive from non-competitive races. For organizations with limited polling budgets, this triage function could concentrate expensive resources on the most consequential contests.

6.5 Cross-Model Robustness

To test whether our results depend on the specific choice of Qwen3-30B-A3B, we replicated the full five-election backtest and the 2025 third-candidate analysis with EXAONE-4.0-32B, a Korean-primary 32-billion parameter language model. This yields a natural cross-model contrast: Qwen3 is multilingual with Chinese primacy, while EXAONE is Korean-primary, and the two models come from independent training pipelines.

Table 15 reports the backtest comparison across all four models we evaluated (Qwen3-30B-A3B, Llama-3.1-8B, EXAONE-4.0-32B, and DeepSeek-R1-8B).

Aggregate accuracy transfers between the two large models within 1–2 percentage points: EXAONE-4.0-32B averages 7.8%p MAE versus Qwen3's 6.5%p. However, the per-election breakdown reveals that the two models fail in different places. Qwen3 achieves 3/3 presidential winner accuracy but 0/2 general winner accuracy; EXAONE reverses this, correctly calling both general elections (2020, 2024) but missing the 2025 presidential winner. Notably, EXAONE's party-proxy general election predictions are *substantially more accurate* than Qwen3's (4.1 and 3.3%p MAE versus 5.3 and 4.4%p), and both predicted winners are correct— suggesting that the general election failure we reported earlier is partly model-specific rather than a fundamental limitation of the party-proxy approach.

Opposite political valences on the 2025 race. The most striking divergence is the 2025 presidential result. The actual outcome was 이재명 (Lee Jae-myung, progressive) 49.4% vs 김문수 (Kim Moon-soo, conservative) 41.2%. Qwen3 predicted Lee 61.3% / Kim 38.7%—a progressive bias of +11.8%p on Lee. EXAONE predicted Lee 41.6% / Kim 58.4%—a conservative bias of +17.2%p on Kim of comparable magnitude. The two models exhibit opposite political valences on the same contest, consistent with known findings that LLMs inherit directional political bias from their training corpora. This opposition is a strong argument for ensemble methods: simply

Table 16: Third-candidate analysis: 2-way vs 3-way 2025 presidential predictions across models.

Candidate	Qwen3-30B		Llama-3.1-8B		DeepSeek-R1-8B		EXAONE-4.0-32B	
	2-way	3-way	2-way	3-way	2-way	3-way	2-way	3-way
이재명 (Lee JM)	61.3	–	72.9	58.6	63.7	33.2	41.6	31.1
김문수 (Kim MS)	38.7	–	27.1	27.6	36.3	35.0	58.4	43.9
이준석 (Lee JS)	–	–	–	13.8	–	31.8	–	25.0
Actual (3-way)	Lee JM 49.4 Kim MS 41.2 Lee JS 8.3							

Table 17: Dynamo-K simulation accuracy vs. final pre-election polls. MAE in percentage points.

Election	Final Poll		Qwen3-30B		Llama-3.1		DS-R1-8B		EXAONE-4.0	
	MAE	Win	MAE	Win	MAE	Win	MAE	Win	MAE	Win
19th Pres. 2017	1.9	✓	13.3	✓	13.1	✓	7.9	✓	12.5	✓
21st Gen. 2020	9.2	✓	5.3	×	10.8	✓	5.0	✓	4.1	✓
20th Pres. 2022	1.9	✓	2.1	✓	13.6	×	20.7	×	6.4	✓
22nd Gen. 2024	10.0	✓	4.4	×	5.8	×	1.8	✓	3.3	✓
21st Pres. 2025	7.3	✓	7.1	✓	18.8	✓	9.6	✓	12.5	×
Average	**6.1**		**6.5**		**12.4**		**9.0**		**7.8**	

averaging Qwen3 and EXAONE predictions for 2025 would yield Lee 51.5% / Kim 48.6%, within 2.1%p of the actual result.

Third-candidate collapse replicates across models. Table 16 compares 2-way and 3-way 2025 predictions across all models with completed 3-way runs. Despite opposite political valences, both Qwen3 and EXAONE exhibit the same qualitative third-candidate dynamic: adding 이준석 (Lee Jun-seok) to the ballot collapses predicted shares toward him far beyond his actual 8.3% support. EXAONE assigns Lee Jun-seok 25.0% of the vote in the 3-way scenario, and vote accounting suggests the shift draws roughly 58%/42% from Kim/Lee respectively —the inverse of Qwen3's draw pattern, but with the same overall effect. This replication across models with opposite political valences supports our earlier interpretation (Sec. 6.2) that the collapse is a general LLM artifact—plausibly hindsight leakage of Lee Jun-seok's public profile—rather than a feature of any single model family.

Comparison to polling. Table 17 places all four models against final pre-election polls. Both Qwen3-30B and EXAONE-4.0-32B achieve average MAE within 2%p of the polling baseline (6.1%p), at a substantially lower marginal cost per scenario (Sec. 6.4).

Summary. Across this cross-model comparison, the architectural findings of Dynamo-K—competitive-with-polling MAE, census-grounded agent viability, and the third-candidate collapse artifact—all reproduce with a second independently trained model. The directional bias on individual contested elections, by contrast, is highly model-dependent, with two similarly-sized models producing +11.8%p and −17.2%p signed errors in opposite directions on the same race. This is a strong empirical argument for ensembling or post-hoc calibration in any production deployment, and it sharpens the limitation discussed next.

6.6 Limitations

Several limitations constrain the generalizability of our findings:

Census temporal mismatch. All simulations use 2020 census data, introducing demographic drift for elections 3–5 years before or after the census year. Demographic changes in age distribution, urbanization, and education levels are not captured.

Static agents. Agents do not update beliefs during a simulated campaign. Real voters respond to debates, scandals, economic shocks, and social network effects that unfold over weeks. Our single-shot simulation captures election-day preferences but not the dynamic process of opinion formation.

Small population. With 5,000 agents across 17 *sido*, each province contains an average of ∼294 agents, with smaller provinces having as few as ∼50. This limits sub-national precision and makes *sido*-level predictions noisy.

Model-dependent directional bias. Our primary results use Qwen3-30B-A3B. The cross-model robustness check with EXAONE-4.0-32B (Sec. 6.5) shows that aggregate accuracy metrics transfer between the two models to within 1–2 percentage points, and that the third-candidate collapse artifact is reproduced. However, the two models exhibit opposite political valences on the 2025 presidential race (Qwen3: +11.8%p progressive bias on Lee Jae-myung; EXAONE: −17.2%p conservative bias on the same candidate), demonstrating that directional bias on contested races is highly model-dependent. Two-model coverage is still narrow, and model-agnostic bias correction—calibrating against a held-out election, or ensembling across models with opposing valences— remains an important open problem.

No social interaction. Agents vote independently without social network effects, peer influence, or information cascading. Real electoral dynamics include bandwagon effects, strategic voting, and social desirability pressures that our model ignores.

Sample of elections. Six elections is a small sample for drawing robust statistical conclusions. The 3/3 presidential accuracy is across only three races (binomial 95% CI [29.2%, 100%]) and could reflect favorable conditions rather than systematic capability; the held-out 2022 local result (Sec. 5.4) is a single out-of-sample data point and addresses, but does not resolve, this sample-size concern. Extending Dynamo-K to additional Korean elections (by-elections, earlier presidentials, local cycles 2018 and 2026) is the most direct path to a tighter bound.

7 Learned Calibration Adapter

The cross-model opposite-valence finding (Sec. 6.5: Qwen3 +11.8%p vs. EXAONE −17.2%p on the 2025 race) motivates a more systematic approach to bias correction than ad-hoc averaging. We introduce the **Orientation × Sido Logistic Reweighting (OSLR)** adapter, a learned generalization of the hand-coded Stage-3b Gallup calibrator (Sec. 3.4). Rather than flipping agents toward a pre-specified 26/48/26 target, OSLR learns per-agent weights from backtest errors:

$$w_i \propto \exp\big(\beta_{m(i),\,o(i)} + \beta_{o(i),\,s(i)}\big), \tag{5}$$

normalized per (election, model) so $\sum_i w_i = N$. Predicted shares are $\hat{v}_c = \sum_i w_i \cdot \mathbb{1}[\text{vote}_i = c] / \sum_i w_i$. Parameters $\beta_{m,o}$ (model × orientation, 4×3) and $\beta_{o,s}$ (orientation × sido, 3×17) are fit by minimizing mean-squared per-candidate share error on training elections, with L2 regularization. Crucially, *candidate names never enter the adapter*—candidates are mapped only to stable features (party orientation, incumbent flag, third-party flag), so OSLR generalizes to unseen candidate lineups.

Table 18: Leave-one-election-out CV of the OSLR calibration adapter, best $\lambda = 0.01$. "Best single" is the lowest-MAE single model per election (oracle). The ensemble mean averages all four models. OSLR applies learned reweighting, then averages across models.

Held-out	Best single	Ens. MAE	Ens. Win	OSLR MAE	OSLR Win
19th Pres. 2017	7.9	11.7	✓	11.8	✓
21st Gen. 2020	4.1	4.7	✓	2.4	✓
20th Pres. 2022	2.1	7.0	×	5.5	×
22nd Gen. 2024	1.8	1.3	✓	4.8	×
21st Pres. 2025	7.1	5.8	✓	4.7	✓
Average	**4.6**	**6.1**	**4/5**	**5.8**	**3/5**

7.1 Leave-One-Election-Out Generalization

We evaluate OSLR with leave-one-election-out (LOO) cross-validation: for each held-out election, the adapter is fit on the other four and then applied (per model, then averaged across models) to the held-out agents. We sweep $\lambda \in \{10^{-3}, 10^{-2}, 5 \times 10^{-2}, 10^{-1}, 0.5, 1, 5\}$ and select $\lambda = 10^{-2}$ as the CV-mean-MAE minimizer. Table 18 compares OSLR against two baselines: the oracle single-model lower bound ("best single") and the simple 4-model ensemble mean.

OSLR substantially improves MAE on three of five held-out elections: -2.33%p on the 2020 general election, -1.49%p on the 2022 presidential, and -1.03%p on the 2025 presidential. It is essentially neutral on the 2017 race ($+0.13$%p) and degrades on the 2024 general by $+3.49$%p, which is also the only held-out case where the simple ensemble already attained near-perfect accuracy (1.27%p) that the adapter could not improve. In aggregate terms, OSLR improves average MAE from 6.1%p to 5.8%p but loses one winner prediction compared to simple ensembling (3/5 vs. 4/5).

7.2 Cold Held-Out Test: 2022 Local Election

The LOO design above leaves a residual question: with only five training points and a 63-parameter adapter, do the gains generalize to a structurally different held-out election that the adapter never had a chance to memorize? To answer this we hold out the 2022 local election entirely. The adapter is fit on the five original elections, then evaluated cold on the 2022 local result (Sec. 5.4). The four-model predictions are aggregated to the named-candidate slate in each of the 17 sido and the per-party vote shares are compared against the certified NEC totals.

Table 19 reports the held-out evaluation alongside the simple ensemble baseline. OSLR reduces national per-party MAE from 2.29%p (simple ensemble) to 1.57%p; both methods tie on sido-hit at 12/17 and both miss the actual winning party, because the four-model split is evenly balanced (Qwen3 and Llama predict 더불어민주당; EXAONE and DeepSeek-R1 predict 국민의힘; see Table 6). The oracle best-single-model (EXAONE-4.0-32B at 0.54%p, 15/17) remains sharper than OSLR, but OSLR is the better unconditional bet when which-model-is-best is not known in advance, and its 0.7%p improvement over the simple ensemble shows the per-agent reweighting transfers cold to a structurally different election.

Even at face value, the experimental design is informative. $\beta_{o,s}$ is *defined* on sido-level variation ($3 \times 17 = 51$ of the 63 adapter parameters), and the 2022 local election is the single race-type that surfaces sido-level signal with no presidential coattail and no constituency-level dilution. If OSLR transfers, the gain is evidence that the adapter has identified real sido-level reweighting structure

Table 19: Cold held-out evaluation of OSLR on the 2022 local election. The adapter is fit on the five original elections with $\lambda = 0.01$ and applied to the held-out agents across the four available models (Qwen3-30B-A3B, Llama-3.1-8B, EXAONE-4.0-32B, DeepSeek-R1-8B); "simple ensemble" is the arithmetic mean of per-party shares across the same four models.

Method	Nat. MAE (%p)	Sido hit
Best single (oracle, EXAONE-4.0-32B)	0.54	15/17
Simple ensemble mean	2.29	12/17
OSLR (trained on 5)	**1.57**	12/17

rather than overfit to the five training elections; if it fails, the regression localizes overfitting to a single sub-population.

7.3 Pooled vs. Per-Model Variants

Fitting a single pooled adapter across all four models outperforms fitting one adapter per model (5.83%p vs. 6.03%p average, both 3/5 winners). The pooled variant benefits from the roughly 4× larger training set and from the cross-model regularization provided by the shared $\beta_{o,s}$ parameters—consistent with the intuition that opposite-valence models are better handled jointly than in isolation.

7.4 Effective Degrees of Freedom and Permutation Test

The nominal parameter count $63 = 4 \times 3 + 3 \times 17$ overstates the adapter's flexibility. Computing the trace of the hat matrix $J(J^\top J + \lambda I)^{-1} J^\top$ where J is the finite-difference Jacobian of predicted shares with respect to β at the fit, the *effective* degrees of freedom under $\lambda = 0.01$ is **7.0** (one-ninth of the nominal count). Under the L2 penalty, the adapter operates closer to a 7-parameter than a 63-parameter model, which directly bounds the overfitting capacity available to it.

We complement this with a permutation test on the held-out 21st-general-2020 fold (the case where OSLR's gain over simple ensembling is largest, −2.33%p). We randomly permute the training-election labels in the actuals dictionary, refit OSLR on each of $N = 100$ permutations, and compare the held-out MAE distribution against the real fit. The real held-out MAE is **2.36%p**; the null distribution has mean 2.80%p and standard deviation 0.99%p, placing the real fit at the 56-th percentile of the null. The real fit is below the null mean but not extreme on this single fold—consistent with the L2 penalty being the dominant overfitting control rather than the labeling itself carrying a strong signal at $N = 100$ permutations on a 5-election training set.[1]

7.5 Interpretation and Use

The pattern of wins and losses is informative. OSLR helps most where simple ensembling already fails (2020: the actual Democratic winner is correctly recovered; 2022: the tight 윤석열/이재명 margin is partially corrected) and hurts most where simple ensembling already works (2024: adapter overfits to training-set general-election patterns that do not fit the 2024 structure). A practical deployment should therefore switch between simple ensembling and OSLR based on a confidence

[1]Full permutation results are stored in `data/calibration/oslr_overfitting_analysis.json` produced by `scripts/oslr_defense_analysis.py`.

signal—for instance, applying OSLR only when cross-model variance is above a threshold, retaining simple averaging when models already agree. We leave this selective calibration to future work. The 2024 failure also highlights the fundamental small-sample challenge: with only five elections to train on, any learned adapter risks overfitting to training-election idiosyncrasies. Extending Dynamo-K to additional Korean elections (local elections, by-elections) or to other East Asian democracies would sharpen this estimate.

8 Conclusion

We have presented Dynamo-K and used it as a diagnostic instrument to characterize Korean-language LLM political behavior across four independently-trained models and six certified Korean elections (2017–2025), with one held out as a cold test for the calibration adapter. Our central claim is not that LLMs can replace polling, but that census-grounded agent simulation surfaces specific, reproducible failure modes that are otherwise hidden behind aggregate-accuracy summaries.

Three mechanism findings are robust across the four models. First, a progressive lean directionally consistent with the GPT-family liberal lean Feng et al. [4] document for English-language LLMs surfaces in Korean voter-decision simulation: moderate agents vote progressive 97% of the time under a vanilla prompt and 82% with demographic conditioning alone, dropping to 59% only under explicit mitigation. Second, third-party candidate collapse decomposes into a salience-failure regime and a decision-bias regime with different per-model signatures, and scenario reframing alone—without architecture changes or hidden labels—recovers 62% of the 2017 MAE. Third, what appears as "conservative bias" in general-election simulation is in fact bidirectional regional polarization collapse, evident only when the error is decomposed by sido. A fourth, model-dependent finding shows that two similarly-sized LLMs produce opposite-direction errors of comparable magnitude on the same race, an empirical case for ensembling and for the OSLR calibration adapter that we evaluate cold on a held-out 2022 local election.

As validation that the diagnostic apparatus is well-calibrated enough to be informative, the system predicts presidential winners 3/3, with a best-case 2.1%p MAE on the 0.73%p-margin 2022 race, and the constituency-level pipeline flips the 2024 general from incorrect to correct (124 vs. 118 seats). The held-out 2022 local election adds a sixth data point that the calibration adapter never had a chance to memorize.

The remaining open problems are sample size and language coverage. Six elections is enough to identify the three mechanism findings but not to bound how they vary across political environments; extending Dynamo-K to additional Korean contests (by-elections, local cycles 2018 and 2026) and to other non-English democracies with available census microdata and political orientation surveys is the most direct way to broaden both the mechanistic and the geographic scope of LLM-based political simulation. The Dynamo-K pipeline is open-source and operates at approximately $0.25 per simulation—a cost structure that, in our framing, enables *diagnosis at scale*, not polling replacement.

A Claude-Opus Audit of Reasoning-Text Mention Rate

Disclosure. The codes reported in this appendix were produced by Claude-Opus (Anthropic's `claude-opus-4-7`), not by a human annotator. We initially described this as a "human validation"; that framing was inaccurate, and we correct it here. The audit is an LLM-against-LLM rubric check, and the limitations of that arrangement (no independent human ground truth; possible self-favoring

Table 20: Claude-Opus audit of regex-derived 안철수 mention rate in the 2017 simulation (50 traces per model, 25 regex-hits + 25 non-hits each). “Genuine” = regex hits that Claude-Opus coded as substantive consideration; “Name-drop” = regex hits coded as list-item enumerations only. All 25 non-hit traces per model were coded as not engaging with 안철수 (true negatives, omitted from the table). “Rationale match” counts mentioned-subset traces whose reasoning Claude-Opus coded as supporting the chosen vote. These are LLM-coded judgments, not human-coded ones.

Model	Genuine	Name-drop	Agreement	Rationale match
Qwen3-30B-A3B	23	2	96%	25/25
Llama-3.1-8B	25	0	100%	23/25
EXAONE-4.0-32B	25	0	100%	24/25
DeepSeek-R1-8B	16	9	82%	25/25
Pooled	89	11	94.5%	97/100

biases from using one LLM to grade the surface features of other LLMs’ outputs) are inherent and should be read into every number in this section.

Table 11 reports a regex-derived mention rate for the 2017 third-party candidate 안철수 within agent reasoning text. To check whether regex hits correspond to substantive consideration rather than list-style name-dropping, we drew a stratified random sample of 200 reasoning traces—50 per model, each split 25/25 between regex-hit and non-hit—and had Claude-Opus code each trace on two axes against a written rubric: (i) does the agent genuinely engage with 안철수 as a viable option (compare his pledges substantively, weigh him in the decision), versus a list-item name-drop or one-line dismissal; and (ii) does the stated rationale support the chosen vote, or does it glow about a non-voted candidate. The rubric, the coded CSV, and the coding script are released at `data/calibration/reasoning_claude_opus_eval_2017.csv` and `scripts/apply_claude_opus_codes.py`.

Two patterns inform interpretation of Table 11, with the caveat that they reflect Claude-Opus’s judgments rather than human ground truth. First, the salience-failure cluster (EXAONE-4.0-32B and Llama-3.1-8B) reads cleanly under Claude-Opus’s coding: when these models mention 안철수, the coder marked the trace as substantive in every sampled case, which is consistent with the low Ment rate in Table 11 being a genuine salience failure (the model simply does not retrieve the candidate in most agents) rather than a regex artifact. Second, the decision-bias cluster splits. Qwen3-30B-A3B’s regex hits are coded as genuine engagements 96% of the time, supporting the decision-bias interpretation as-stated. DeepSeek-R1-8B’s regex hits are coded as list-style enumerations of all candidates without substantive weighing in 9 of 25 sampled cases (82% agreement), so the raw regex mention rate (22.0% in Table 11) appears to overstate DeepSeek’s actual engagement. Restricted to the genuinely-engaged subset under Claude-Opus’s coding, DeepSeek’s sample conditional vote rate is $9/16 = 56\%$, considerably higher than the regex $V|M = 34.5\%$ reported in Table 11—under this audit DeepSeek looks closer to the EXAONE pattern (high $V|M$ among coded considerers), with its distinctive feature being list-style enumeration rather than a uniquely strong decision bias. Rationale-vote consistency is high across models (97/100 sampled traces); the three mismatches Claude-Opus flagged (two Llama, one EXAONE) involve agents whose reasoning text is more enthusiastic about 안철수 than about their actual vote (typically 문재인), a pattern consistent with the LLM progressive-bias mechanism diagnosed in Section 6.1.

Limitations of this audit. Because every code in this appendix was produced by a single LLM (Claude-Opus) and not by a human, the agreement and rationale-match numbers measure consistency between two LLMs’ surface treatments of the same reasoning, not correctness against an external ground truth. A genuine human validation—ideally with multiple annotators and inter-rater agreement—remains future work. We retain this appendix because the rubric-coded numbers materially change the interpretation of the DeepSeek-R1-8B row of Table 11 relative to the raw regex baseline, and we judged that flagging that interpretive shift, with full disclosure of how the codes were obtained, is preferable to omitting the audit entirely.

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